

The dataset contains 1,200 records and 13 features with no missing values or duplicate records, making it suitable for machine learning analysis.

Teenagers spend an average of 4.5 hours per day on social media.

The scatter plot indicates that higher social media usage is generally associated with lower sleep duration, suggesting a potential negative impact on sleep habits.

Most teenagers sleep between 5 and 8 hours.

Lower sleep duration appears to be associated with higher depression risk.

The dataset is highly imbalanced, with far fewer depressed cases than non-depressed cases.

This explains why overall accuracy is very high while the number of positive (depressed) cases is small.

The result which we got through these different ML Models:

Model	Accuracy
Logistic Regression	98.75%
Decision Tree	99.17%
Random Forest	97.92%

The confusion matrix indicates that most students were classified correctly.

The ROC curve demonstrates strong classification performance, indicating good separation between depressed and non-depressed classes

Business Analysis:

A) Schools should use indicators such as:

High stress level, High anxiety level, High social media addiction

B) Limit daily social media usage. Reduce screen time before bedtime.

C) Take regular breaks from digital devices.